

# Santhosh Narayanan Baburaman

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## EDUCATION

**M.S. Analytics**, University of Southern California (GPA: 3.92/4.0) Jan 2025 – Dec 2026  
**B.Tech. Computer Science and Engineering**, VIT (CGPA: 8.41/10) Sep 2020 – Apr 2024  
*Publication:* Co-authored an IEEE conference paper on predictive analytics for transportation, modeling travel time, fare, and traffic speed across 1M+ NYC taxi trip records

## SKILLS

- **Data Engineering:** Python, SQL, dbt, ELT/ETL, Dimensional Modeling, SCD Type 2, Data Quality Testing, Git
- **Platforms & Ops:** Snowflake, BigQuery, PostgreSQL, Redshift, GCP (Vertex AI, Kubeflow), GitHub Actions
- **BI & Reporting:** Power BI (DAX, Power Query), Tableau, Alteryx, Streamlit, Excel

## EXPERIENCE

**Data Analyst Intern** | Philadelphia Department of Aviation Jun 2026 – Aug 2026

- Owned 2 production data pipelines covering ~30% of airport operating revenue, replacing ~60 hours/month of manual review with automated, reusable workflows
- Developed a Python → Parquet → Alteryx → Power BI pipeline over 800K+ rows of flight-level billing data spanning 4 fiscal years; standardized airline, aircraft, and timestamp fields with fallback logic for missing or malformed source values
- Restructured ground transportation reporting on PropWorks SQL → Amazon Redshift, correcting root-cause aggregation-order and duplicate-rate defects, with fiscal-year time intelligence and anomaly detection in DAX
- Integrated both pipelines to compute cross-domain passenger capture rates, and documented each workflow so the Finance team can run it independently every reporting cycle

**IT Student Assistant** | USC Office of the Provost Jun 2025 – Present

- Provide IT support across 15+ administrative departments, configuring and reimaging devices for new users

**Founder & President** | ISE Analytics Student Association (IASA), USC Feb 2025 – May 2026

- Founded USC's first analytics student organization, hosting 12+ events for 500+ students and 20+ professionals

**AI Intern** | SmartInternz May 2023 – Jul 2023

- Led a 3-person team building a CNN-LSTM model that generates captions for 8,000 Flickr8k images in Python

**Azure Cloud Intern** | SmartKnower Sep 2022 – Oct 2022

- Deployed and managed virtual machines and containerized applications on Microsoft Azure cloud infrastructure

## PROJECTS

**Consulting Firm Data Warehouse** (dbt, Snowflake, SQL, Python)

- Constructed a multi-layer ELT pipeline in dbt on Snowflake, transforming 17 normalized source tables through 17 staging models into 3 subject-area datamarts for a 4-person team
- Resolved inconsistent project-type values, reserved-word conflicts, and missing temporal columns in staging via SQL window functions (LEAD)
- Architected 7 conformed dimensions including an SCD Type 2 dimension preserving salary and title history, plus a generated date spine for consistent time aggregation
- Authored 39 automated dbt tests (unique, not null, accepted values) and proactively reconciled design docs against the production schema, documenting 13 discrepancies and unblocking 3 teammates' fact-table work within 24 hours

**Sleep & Cognition Research** | Student Researcher, Suen Lab, USC (Python, BigQuery) Sep 2026 – Present

- Built an 11-step cleaning pipeline for 6 years (2021–2026) of Fitbit sleep logs across 5 source tables in BigQuery: overlap resolution, DST correction, cross-source deduplication, and one main sleep per person-night
- Reconstructed fragmented sleep into continuous episodes (gap stitching within a night window) to build a second sleep dataset, and ran every analysis on both versions to test robustness

**Automated Sales Analytics Copilot** (Python, PostgreSQL, GitHub Actions, Streamlit) 🐙 GitHub

- Designed a PostgreSQL star schema with 5 KPI views tracking revenue, margin, and discount impact across a 10K-order, \$2.3M dataset
- Automated nightly inference and anomaly alerting through a scheduled GitHub Actions pipeline, deploying a full-stack Streamlit app with Slack and email notifications